FINAL PROJECT

202322621

Individual Assignment

Phase 2

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1. Stakeholder Analysis and Conflict Resolution

Stakeholder Identification

Following the viewpoint-oriented approach to requirements elicitation, the system's stakeholders are classified below by viewpoint type, which structures what kind of information each stakeholder can be expected to supply.

Stakeholder	Viewpoint type	Interest in the system
Student	Receiver of services	Views enrolled courses and assignments; submits and resubmits work before deadline; views released marks and feedback
Lecturer	Receiver of services	Creates assignments; grants individual extensions; marks submissions; writes feedback
Head of Department	Receiver of services	Reviews and moderates marks across the department; releases approved marks
Administrator	Receiver of services	Manages accounts, departments and courses; assigns lecturers; enrolls students
Principal	Indirect stakeholder (sets organizational policy)	Institutional accountability; requires the system to be simple for non-specialist staff and to protect unreleased marks
IT Officer	Data source/sink + representation framework	Operational constraints on the technical environment (offline capability, performance, password protection)
Student representative	Receiver of services (collective voice)	Advocates for fairness in resubmission and late-work handling on behalf of the student body

Conflicts and Ambiguities in the Stakeholder Statements

#	Conflicting statements	Nature of conflict	Resolution	Reflected in
C1	Principal: “students must not see any mark until it has been checked by the Head of Department.” vs. Lecturer: wants “students to get their marks as soon as I have marked the work.”	Direct contradiction on when a mark becomes visible to the student.	The Principal’s requirement is authoritative (institutional policy over individual preference). The Lecturer’s underlying need is preserved by separating mark from feedback: written feedback is visible immediately after marking; the numeric mark stays hidden until HOD moderation and release.	Submission life cycle (Section 5.3) — “marked” and “released” as distinct states
C2	HOD: precise penalty rule (10%/day late, capped at 3 days, then zero). vs Student representative: “late work should not simply be rejected.”	Not strictly contradictory, but the Student representative’s statement is ambiguous: unclear if it objects to a penalty existing, or only to outright non-acceptance.	The HOD’s quantified rule is adopted as the authoritative penalty policy, since it is the only statement already verifiable as written. The Student representative’s statement is interpreted as consistent with it: a graduated penalty is not “simple rejection,” and the system must accept late submissions (even beyond the 3-day window, at zero marks) rather than block them.	Late-penalty business rule (SRS, Section 4.2)
C3	Lecturer: grants extensions to individual students. vs. (no stakeholder statement addresses class-wide extension at this stage).	Not a conflict: a scope boundary. Class-wide extension appears only later, as CR1 (Phase 8).	The Phase 2–4 system supports individual extensions only. Class-wide extension is deliberately out of scope for v1.0, to be added under change control rather than anticipated speculatively.	Use case “Grant extension” (individual scope only); CR1 impact analysis (Phase 8)
C4	Principal: “the system must be simple” (for non-IT-expert teachers).	Not a conflict: an unverifiable goal, not a requirement, per the lecture’s goal-vs. requirement distinction.	Restated as a verifiable NFR: a bounded number of steps and a maximum training time for the Lecturer’s core workflow (creating an assignment, marking a submission), so “simple” becomes testable.	SRS non-functional requirements (Section 4.2)

Open Items Flagged for Gate 1 Validation

#	Item	Why it's flagged	Provisional assumption
O1	Who may view the audit log	The case study requires the school to see "who changed a mark, when, and why," but names no role with access rights.	Administrator and HOD only, pending confirmation
O2	Resubmission after the original deadline	The Student representative supports resubmission "before the deadline"; it's unclear whether resubmission is still allowed once work is marked late (within the 3-day window).	Resubmission permitted only before the original deadline, pending clarification

2. Software Requirements Specification

Functional Requirements

ID	Requirement	Priority	Verification / Metric	Resolves
FR01	The system shall allow the Administrator to create, edit and deactivate user accounts for all four roles.	Must	Test: account created/edited/deactivated and reflected immediately in login behaviour	—
FR02	The system shall allow the Administrator to create and manage departments and courses.	Must	Test: department/course record created, editable, queryable	—
FR03	The system shall allow the Administrator to assign a Lecturer to a course.	Must	Test: assignment reflected in the Lecturer's course list	—
FR04	The system shall allow the Administrator to enroll a Student in a course.	Must	Test: enrolment reflected in the Student's course list	—
FR05	The system shall allow a Lecturer to create an assignment, specifying a deadline and a maximum mark, for a course they are assigned to.	Must	Test: assignment visible to enrolled students with correct deadline/max mark	—
FR06	The system shall allow a Lecturer to grant an individual deadline extension to a named student for a named assignment.	Must	Test: extended student may submit after the original deadline; other students cannot	—

ID	Requirement	Priority	Verification / Metric	Resolves
FR07	The system shall allow a Lecturer to record a numeric mark for a submission, not exceeding the assignment's maximum mark.	Must	Test: mark outside valid range is rejected	—
FR08	The system shall allow a Lecturer to write feedback text accompanying a recorded mark.	Must	Test: feedback stored and linked to the correct submission	—
FR09	The system shall make a Lecturer's written feedback visible to the student immediately after marking, independent of whether the mark itself has been released.	Should	Test: feedback visible pre-release; mark remains hidden	C1
FR10	The system shall allow a Head of Department to review all marks recorded for courses in their department prior to release.	Must	Test: HoD view lists every unreleased mark in-department	—
FR11	The system shall allow a Head of Department to moderate (adjust) a mark before release, requiring a recorded reason for any change.	Must	Test: adjustment rejected if no reason supplied	—
FR12	The system shall allow a Head of Department to release approved marks, making them visible to the relevant students.	Must	Test: mark visible to student only after this action	—
FR13	The system shall prevent a Student from viewing any mark that has not been released, regardless of marking or moderation status.	Must	Test: unreleased mark absent from student view under all states	C1
FR14	The system shall allow a Student to view the list of courses they are enrolled in.	Must	Test: list matches Administrator enrolment records	—
FR15	The system shall allow a Student to view assignments for their enrolled courses, including deadline and maximum mark.	Must	Test: displayed values match Lecturer-entered values	—
FR16	The system shall allow a Student to submit work for an assignment at any time up to its deadline (as individually extended, if applicable).	Must	Test: submission accepted before deadline/extension; time stamped correctly	—
FR17	The system shall allow a Student to resubmit work for an assignment, replacing the prior submission, at any time up to the original deadline.	Must	Test: resubmission overwrites prior file; resubmission after deadline is rejected	O2 (provisional)

ID	Requirement	Priority	Verification / Metric	Resolves
FR18	The system shall automatically deduct 10% of an assignment's maximum mark for each full day a submission is late, up to three days, and shall record a mark of zero for submissions more than three days late.	Must	Test: mark deduction calculated correctly at 1, 2, 3 and 4+ days late	C2
FR19	The system shall allow a Student to view their released marks and accompanying feedback for each submitted assignment.	Must	Test: values match what the HoD released	—
FR20	The system shall require every user to authenticate with a username and password before accessing any role-specific function.	Must	Test: unauthenticated access attempt is rejected	—
FR21	The system shall restrict each authenticated user to only the actions permitted for their assigned role.	Must	Test: role-inappropriate action attempt is rejected for each of the four roles	—
FR22	The system shall record every change to a mark in an audit log, capturing who made the change, when, the old value, the new value, and a reason.	Must	Test: log entry created and complete for every mark-changing action	—
FR23	The system shall allow a Lecturer to view their own created assignments, filterable by deadline.	Could	Test: filtered list matches expected subset	—

Non-Functional Requirements

ID	Requirement	Category	Priority	Verification / Metric	Resolves
NFR01	The system shall complete user login in under 2 seconds with the full dataset of 800 students stored.	Product — Efficiency	Must	Measured login time under load test with 800 student records	IT Officer statement
NFR02	A Lecturer with no prior training shall be able to create an assignment and mark a submission within 10 minutes of first use, consulting only in-app guidance.	Product — Usability	Must	Timed usability test with an untrained participant	C4 (Principal "simple")

ID	Requirement	Category	Priority	Verification / Metric	Resolves
NFR03	The system shall remain fully operational with no internet connectivity, performing all core functions using only local storage.	Product — Dependability	Must	Test: all core functions exercised with network disabled	IT Officer statement
NFR04	User passwords shall never be stored in plain text; the system shall store only salted cryptographic hashes generated using a recognized library.	Product — Security	Must	Code review + database inspection confirms no plaintext password field	IT Officer statement
NFR05	A mark-change audit record shall be retrievable in response to a query within 3 seconds, for records up to five years old.	Organizational — Supportability	Should	Measured query response time against a populated five-year log	Case study: audit requirement
NFR06	Access to a learner's personal academic records shall be restricted to users whose role permits it, consistent with applicable data protection obligations.	External — Regulatory	Must	Test: role-inappropriate access attempt is rejected; reviewed against Section 11 (PLE issues)	Feasibility study, Section 3.1

3. Use Case Model

